Chicago Food Inspections - Phase One

Team111
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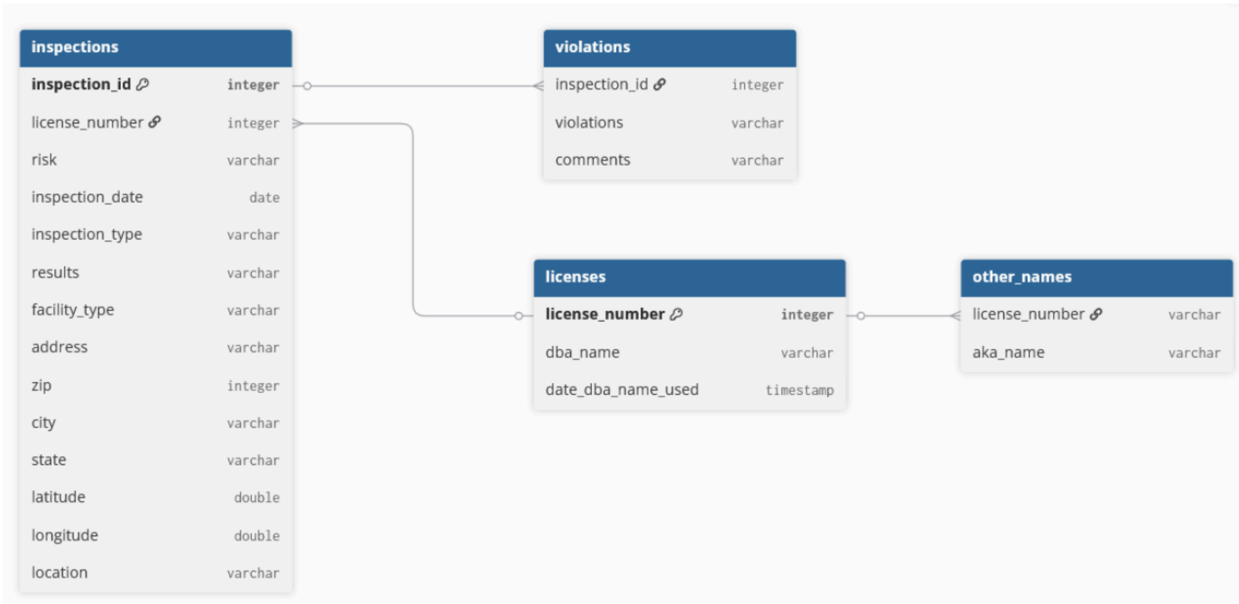
1. Describe the Dataset

1a. Database Schema



1b. Description of the Origin of the Data

This Chicago Food Inspection data is the results of a series of inspections of various Chicago establishments from 1/4/2010 - 10/22/2025 sourced from the . As per the City of Chicago dataset description, the data columns are originally defined as below. Please note that the dataset description noted that some fields should be filled with specific values, but in the actual dataset, they are populated with values outside of these enumerated values.

- **Inspection ID (Number):** This is not explicitly stated in the dataset description, but it is a unique Inspection ID for each inspection that takes place
- **DBA Name (Text):** Doing Business As (DBA) Name is the legal name of the establishment
- **AKA Name (Text):** Also Known As (AKA) Name that the business also operates under
- **License # (Number):** The License ID is a unique number assigned to the establishment for the purpose of licensing by the Department of Business Affairs and Consumer Protection
- **Facility Type (Text):** Each has one of the following categorizations: Bakery, Banquet Hall, Candy Store, Caterer, Coffee Shop, Day Care Center (for ages less than 2), Day Care Center (for ages less 2 - 6), Day Care Center (combo, for ages less than 2 and 2 - 6 combined), Gas Station, Golden Diner, Grocery Store, Hospital, Long Term Care Center (Nursing Home), Liquor Store, Mobile Food Dispenser, Restaurant, Paleteria, School, Shelter, Tavern, Social Club, Wholesaler, or Wrigley Field Rooftop.

- **Risk (Text):** The level of risk is defined as the risk of adversely affecting the public's health. 1 is the highest risk and 3 being the lowest and the inspection frequency correlates to how high the risk is deemed to be.
- **Address (Text):** The street address of the establishment
- **City (Text):** The city the establishment operates in
- **State (Text):** The state the establishment operates in
- **Zip (Number):** The zip code the establishment operates in
- **Inspection Date (Timestamp):** When the inspection actually occured
- **Inspection Type (Text):** What type of inspection was done
    - Canvass: Common type of inspection with frequency correlated to risk level
    - Consultation: An inspection done at the request of the owner
    - Complaint: An inspection done in response to a complaint
    - License: An inspection done as a requirement for the establishment to receive its license
    - Suspect Food Poisoning: Inspection due to complaints about getting ill due to establishment
    - Task-Force Inspection: Inspection of bar or tavern
    - Re-inspection can occur for any of the above reasons
- **Results (Text):** The findings from the inspection
    - Pass: No critical or serious violations
    - Pass With Conditions: Violations were found but were corrected at time of inspection
    - Fail: Violations were found and were not correctable at time of inspection
    - Establishments that are out of business or could not be located are indicated
- **Violations (Text):** The violations that were found during the inspection
- **Latitude (Number):** Latitude of the place of business
- **Longitude (Number):** Longitude of the place of business
- **Location (Text):** A concatenation of the Latitude and Longitude data fields

In our database schema, the data is broken up into inspections, violations, licenses, and other_names tables.

- The **inspections table** holds the facts of the inspection on where and when it was conducted as well as the result.
- The **violations table** is linked to the inspection table and has the various violations associated with a single inspection. These needed to be broken out from the inspection table since it was originally a "|" delimited value holding multiple violations and commentary on the violation within it. This format with the broken out violations and the separated comments allows for easier querying by violation type.
- The **licenses table** contains just the license number, the DBA Name of the establishment, and the most recent inspection date that the name was used. As license number is stated on the City of Chicago website to be a unique identifier, this has been broken out along with the license number's most recent legal name which will be pulled from their most recent inspection. The confirmed last date the DBA Name was used will be tracked as well to easily compare if there is a more recent name in a row of data that

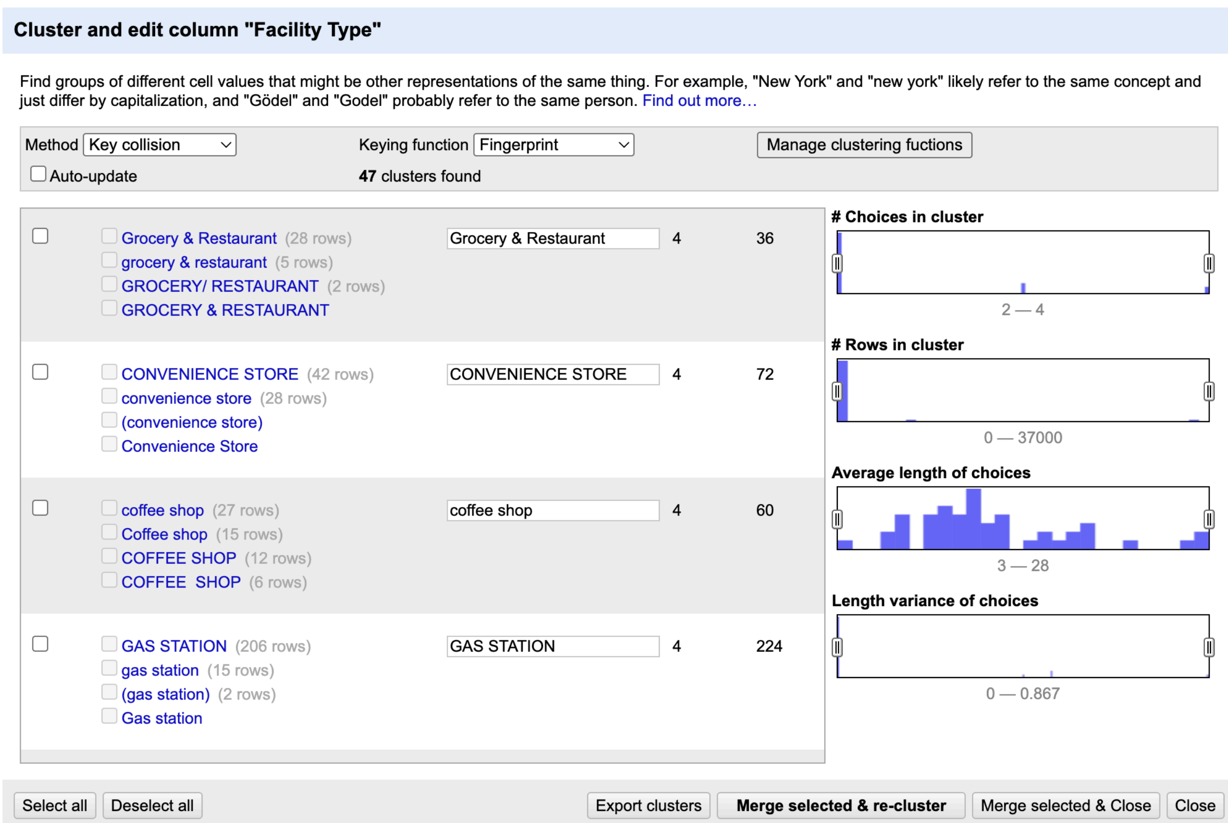
would cause the DBA Name to be updated. The reason the current address information wasn't broken out as well is because establishments can move so it is not unique to a license #, and it seemed material to the inspection at that point in time.

- The **other_names table** contains information on names (AKA Names) other than the establishments' legal name they operate under

2. Use Cases

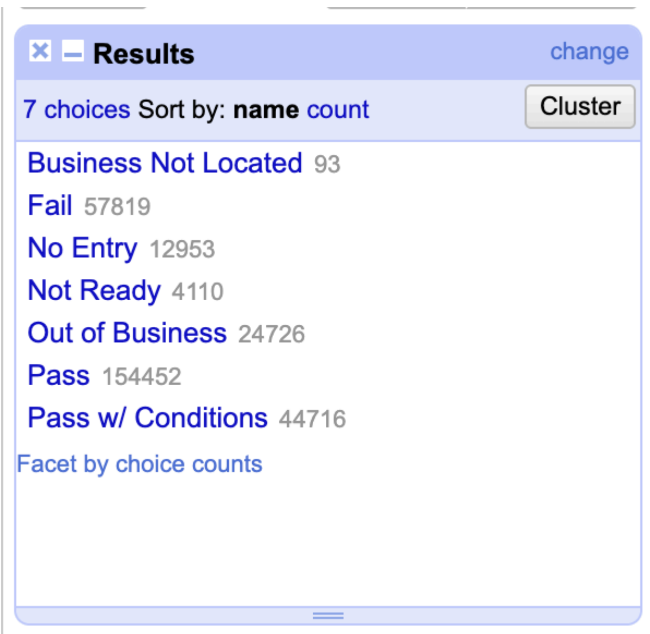
2a. Target Use Case

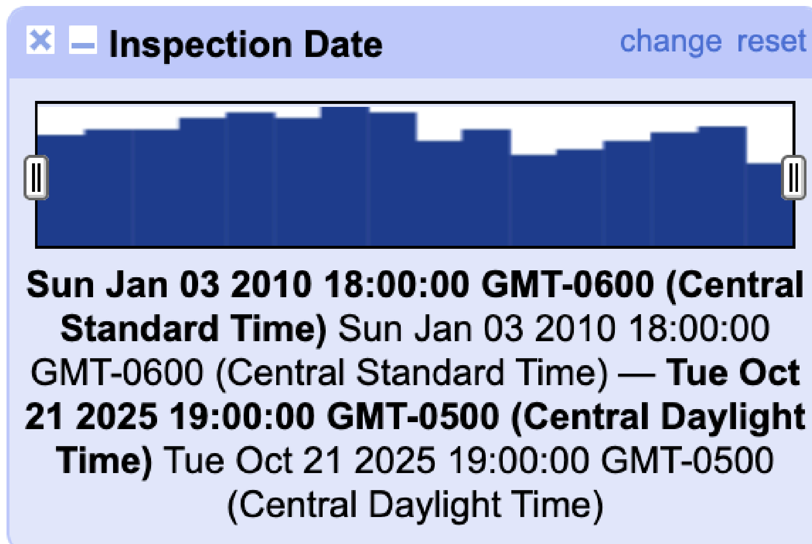
One main use case, U1, that would be interesting to answer from this data set is: which facility types have the highest inspection failure rates after the 7/1/2018 change in definitions, and do some facility types exhibit higher rates of violations? It would be interesting to see how different categories of businesses ranging from coffee shops to daycares or restaurants and grocery stores compare. This might lead to interesting discoveries in the difficulty of standards for different establishments or a trend in business cleanliness in the city. This could be accomplished by grouping by 'Facility Type' and calculating the percentage of failures for each group. However, there are currently multiple facility types that appear to be different, when they likely are not intended to be. For example, in the image below, we can see "Coffee Shop", "coffee shop", "COFFEE SHOP", and "COFFEE SHOP" as 3 different types. Once the facility type naming scheme is normalized, the answer to the above query should be able to be derived.



2b. Zero Data Cleaning Use Care

Other use cases, U0, rely upon a clean subset of the data. One interesting question to answer would be what the percentage breakdown of different results is after the change in definition of violations that occurred on 7/1/2018. For example, what percent of inspections obtain a pass, fail, are not ready, etc. We can see in the image below that this category appears much cleaner than the one above. There aren't different spellings and every row has a value in this column. From a quick review, it also appears that the inspection date category is well formatted and presents a timeline from 2010-2025 which we could reduce down to after the changes in definition were made. It would be interesting to add this column into the query as well and look at whether the composition of results changes year over year or if certain months see higher rates of failures. This could show overall trends in the city's businesses, health code practices, and more.





2c. Never Enough Use Case

Finally, a use case, U2, for which this data is never sufficient would be answering questions such as do different inspectors have different levels of leniency in their assessments or, if the inspections that were triggered by a complaint, how credible was the complaint to begin with? These questions would require more data. For example a column with an inspector name or ID would help answer the first, while a field for the number of complaints filed or trends in foodborne illnesses in the immediate area might help answer the second. The main takeaway is that no amount of data cleaning can provide context beyond what is in the data!

3. Data Quality Problems

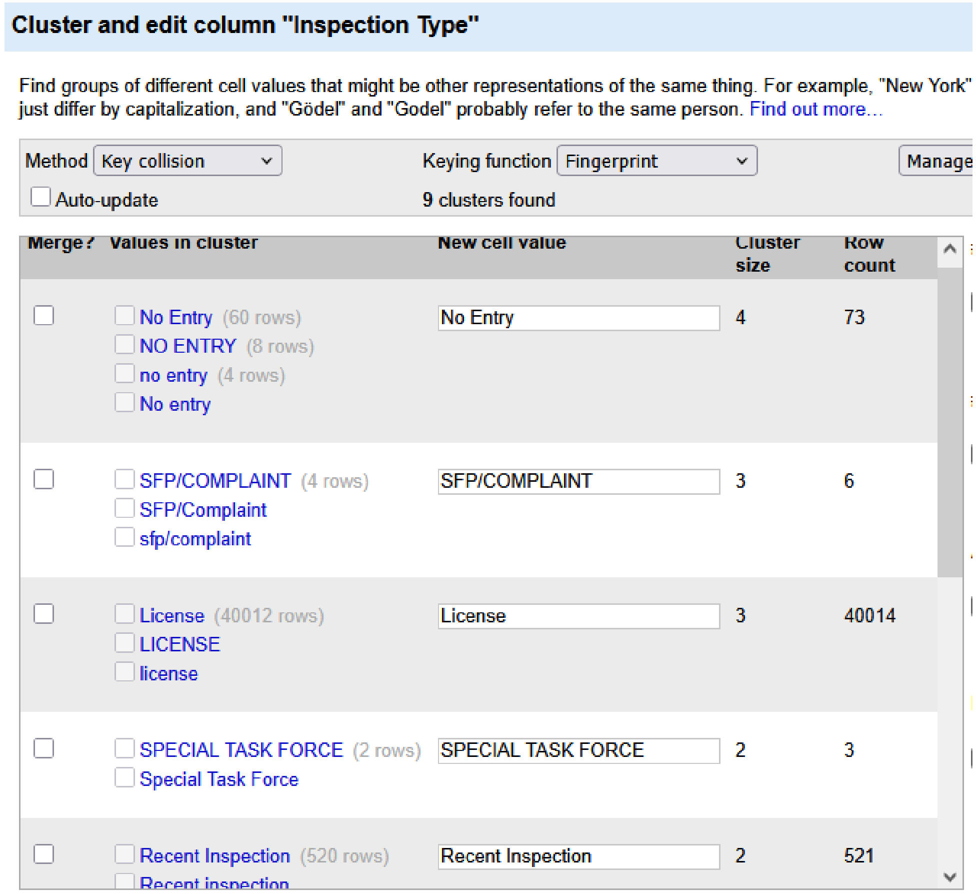
1. Facilities without any “Facility Type”. Since facility type is the main category used to compare inspection failure rates, it would be difficult to accurately measure and compare trends in regards to passing and failing between categories if there were missing values.

All	Inspection ID	DBA Name	AKA Name	License #	Facility Type	Risk	Address	City	State	Zip	Inspection Date	Inspection Type	Results
22.	2625711	WINSTON & STRAWN LLP	WINSTON & STRAWN LLP	3051688		Risk 1 (High)	300 N LA SALLE ST	CHICAGO	IL	60654	10/21/2025	License	Not Ready
43.	2625649	THE POINT NIGHTCLUB	THE POINT VENUE	0			1565 N MILWAUKEE AVE	CHICAGO	IL	60622	10/20/2025	Complaint	No Entry
49.	2625652	SHELL FUEL & MINI MART		3056117		All	7100-7118 S HALSTED ST	CHICAGO	IL	60621	10/20/2025	License	Not Ready
50.	2625635	7-ELEVEN #33753H	7-ELEVEN	3056124		Risk 2 (Medium)	5320 N CUMBERLAND AVE	CHICAGO	IL	60656	10/20/2025	License	Not Ready
132.	2625451	D'S ROTI & TRINI CUISINE	D'S ROTI & TRINI CUISINE	3046732		All	6239 S ASHLAND AVE	CHICAGO	IL	60636	10/16/2025	License	Not Ready
134.	2625474	HAPPY MANNIU SWEET & SPICY INC.	HAPPY MANNIU SWEET & SPICY INC.	3046811		Risk 1 (High)	2601 S HALSTED ST	CHICAGO	IL	60608	10/16/2025	Canvass	Out of Business
183.	2625473	LE SHRMIP	LE SHRMIP	3051585		Risk 1 (High)	2101 S CHINA PL	CHICAGO	IL	60616	10/16/2025	License	Not Ready
218.	2625402	CITGO	CITGO	3041622		Risk 3 (Low)	4759 S CICERO AVE	CHICAGO	IL	60632	10/15/2025	License	Not Ready
286.	2625259	TOP CAKE	TOP CAKE	3046730		All	206 N HOMAN AVE	CHICAGO	IL	60624	10/14/2025	License	No Entry
299.	2625292	"GATEWAY NEWS STAND, INC."	GATEWAY NEWS STAND	3052080		Risk 3 (Low)	1 S WACKER DR	CHICAGO	IL	60606	10/14/2025	License	Not Ready

2. Failed inspections but without any violations. The violations an establishment has can help potential analysis into which facility types fail, what violations are commonly associated with these failures, and how many violations there are on failures.

Results	Violations
Fail	
Fail	
Fail	
Fail	
Fail	
Fail	
Fail	
Fail	
Fail	
Fail	
Fail	
Fail	

3. Inconsistent inspection type value. Possibly the least significant data quality issue, but can still potentially split records and cause inaccurate counting/analysis.



4. As noted above in section 2A of this report, the facility type naming scheme is also not standardized. Notably, instances such as “COFFEE SHOP” and “COFFEE SHOP” would need to be consolidated. For easy consolidations such as this, we can remove extra spaces to standardize the Facility Types to a certain extent. However, a complete standardization to the intended Facility Types as per the City of Chicago list of Facility Types would likely not be possible. This is due to being unable to determine how the City of Chicago would group some of the Facility Type names in the dataset such as “Convenience Store” which was seen in the screenshot in section 2A. However, an analysis of the Facility Types is still possible with the extra Facility Types which cannot be consolidated.

4. Devise an initial plan

S1: Review and Refine Project Definitions

- We will re-verify that the dataset fulfills our U1 analysis. This includes verifying that we can split the violations and then correctly reference them back to the Inspection ID.

S2: Profile the Dataset

- Methodology: We can use Python to get an initial idea of distributions within the Facility Type, Inspection Type, Inspection Date, and Violations.
- Goal: We'll want to quantify the amount of nulls that exist across the columns, leading/trailing/extra whitespace anomalies (e.g. "COFFEE SHOP" vs "COFFEE SHOP") and capitalization differences. We can also do an initial search of the Violations to determine how it is delimited and how to best approach separating them.

S3: Performing Data Cleaning

1. Text Standardization: Import into OpenRefine to merge alike strings within the Facility Type Field
2. Splitting the Data: Using a Python script, we can break apart the Violations column and then break it into a separate normalized table with the inspection ID.
3. Handle Anomalies: Use a Python script to assign values to any null Facility Types. This could be trying to see if a license has a separate inspection with a referenced Facility Type that can be used to populate the value or deciding to bucket it into the "Other" categorization.

S4: Checking Data Quality Improvement

- Ensure the absence of leading/trailing/extra whitespace duplicate Facility Types
- Ensure that null Facility Type values have either been assigned a value by retrieving it from a different line or bucketing it into "Other"

S5: Document and Quantify Change

- We can export the OpenRefine history to record changes made
- Document the changes and total number of Facility Types

Team Member	Phase-II Responsibility	Tools	Deadline
Justin Hughes	S3: Splitting the Data S3: Handle Anomalies	Python	7/22/2026
Stav Dvir	S2: Profile the Dataset S3: Text Standardization	OpenRefine	7/18/2026
Jason Liang	S4: Checking Data Quality	Python	7/25/2026
All Members	S1: Review and Refine Project Definitions	Python	7/13/2026
All Members	S5: Documentation	YesWorkflow	8/1/2026

Bibliography

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Food Inspection Violations Changes. Wayback Machine. (n.d.). https://web.archive.org/web/20190207225953/http://dev.cityofchicago.org/open%20data/data%20portal/2018/06/29/food-violations-changes.html